Chemistry - Graduate Level Assessment 16 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Use the van der Waals parameters for chlorine to calculate approximate values of the radius of a Cl_2 molecule regarded as a sphere.
   (a) 0.185 nm
   (b) 0.110 nm
   (c) 0.125 nm
   (d) 0.200 nm
   (e) 0.095 nm

2. An electron was accelerated through a potential difference of 1.00 ± 0.01 kV. What is the uncertainty of the position of the electron along its line of flight?
   (a) 6.75×10^-10m
   (a) 8.22×10^-11m
   (a) 2.15×10^-12m
   (a) 1.885×10^7 m s$^-1$
   (a) 1.22×10^-9m

3. Express the van der Waals parameters $b = 0.0226\text{dm}^3$ mol^{-1} in SI base units.
   (a) $22.6 \times 10^{\{} - 3\}$ m^3 mol^{-1}
   (b) 0.0226 m^3 mol^{-1}
   (c) $2.26 \times 10^{\{} - 6\}$ m^3 mol^{-1}
   (d) $226 \times 10^{\{} - 6\}$ m^3 mol^{-1}
   (e) $2.26 \times 10^{\{} - 2\}$ m^3 mol^{-1}

4. Of what importance are proteins to biological systems?
   (a) Proteins determine the rate of photosynthesis in plants.
   (b) Proteins are used for insulation in cold environments
   (c) Proteins are the main constituent of cell walls in bacteria.
   (d) Proteins are responsible for coloration of organisms
   (e) Proteins serve as structural material and biological regulators.

5. The weight percent of sodium hydroxide dissolved in water is 50%. What is the mole fraction of sodium hydroxide?

(a) 0.311

(b) 0.5

(c) 1.00

(d) 0.05

(e) 0.10

True / False (10 marks)

Answer True or False

6. True or False: The addition of NaOH to HCl results in a neutral pH of 7 due to complete neutralization of the acid.

7. True or False: A_3.00 liter solution of 0.500 N sulfuric acid contains 102.5 grams of H_2SO_4.

8. True or False: The infrared absorption frequency of deuterium chloride (DCl) shifts from hydrogen chloride (HCl) due to differences in their reduced mass.

9. True or False: The maximum kinetic energy of the $^{-} particle emitted in the radioactive decay of$ He^6 is 7.51 MeV.

10. True or False: The mole fraction of H_2 SO_4 in the solution is 0.1, while the mole fraction of H_2 O is 0.9.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the work done when a gas in a 2.0-liter container at 2.4 atmospheres expands against an external pressure of 0.80 atmosphere. Discuss the implications of this work in thermodynamic terms.

12. Discuss the relationship between pH, pKa, and chemical shift in a solution of sodium phosphate, particularly focusing on the expected chemical shift at the pKa of H_2PO_4.

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